Madalena Osório de Barros

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github.com/madalenarb | madalenarb.github.io/my-cv-portfolio

SUMMARY

Assistant Engineer at NTT DATA (Digital Architecture), integrating SAP data into a Unified Namespace for industrial operations. MSc in Electrical and Computer Engineering at IST (Control, Robotics and AI), with the thesis on GPU-accelerated quantum circuit simulation submitted and the defence pending. Works across AI and data (RAG, LLMs, computer vision) and HPC (CUDA, MPI, profiling), with a strong interest in AI agents and process automation.

EXPERIENCE

NTT DATA June 2026 – Present
Assistant Engineer, Digital Architecture Lisbon, Portugal
- Connect SAP (IDoc) data to a Unified Namespace (UNS) to support an MES rollout for a multinational manufacturer.
- Configure webhooks and event flows in HighByte Intelligence Hub, and validate message mapping and delivery.
- Set up a Docker Compose integration environment for reproducible local testing and faster team onboarding.

TripGoGo (start-up) Nov. 2025 – March 2026
AI & Prompt Engineering Intern Amsterdam, Netherlands
- Designed RAG workflows that ground LLM answers (OpenAI, Gemini) in verified place data, reducing hallucinations.
- Built QA pipelines that validate POIs, coordinates and opening hours against official sources, guiding prompt fixes.

KPMG Portugal July 2024
Data and Analytics Summer Intern Évora, Portugal
- Designed (Figma) and built the React + TypeScript front end of an internal Generative AI chatbot.

EDUCATION

Instituto Superior Técnico (IST), University of Lisbon 2024 – 2026
MSc in Electrical and Computer Engineering Lisbon, Portugal
- **Specialisation:** Control, Robotics and Artificial Intelligence. **Minor:** Contemporary Physics.
- **Thesis:** Accelerating Quantum Circuit Simulation with NVIDIA GPUs (submitted October 2026, defence pending).

Chalmers University of Technology 2024 – 2025
Erasmus Exchange, MSc in Electrical and Computer Engineering Gothenburg, Sweden
- **Focus:** AI & Machine Learning, Mechatronic Systems, Quantum Mechanics.

Instituto Superior Técnico (IST), University of Lisbon 2020 – 2024
BSc in Electrical and Computer Engineering Lisbon, Portugal
- **Academic Merit Diploma** (2022/2023).

PROJECTS

Accelerating Quantum Circuit Simulation (MSc thesis) | Python, C/C++, CUDA, MPI/OpenMP 2024 – 2026
- Benchmarked six state-vector simulators (Qiskit Aer, CUDA-Q, Qulacs, mpiQulacs, Qibo, QuEST) from one CPU thread to 64 nodes and 16 NVIDIA A100 GPUs on the EuroHPC supercomputer Deucalion.
- Built a modular benchmarking framework, explained the results with roofline analysis and top-down profiling (Intel VTune), and turned them into a simulator-selection framework with a prototype tool.

Building Footprint Extraction from Aerial Images | Python, YOLOv8, computer vision 2024
- YOLOv8n-seg instance segmentation for the IEEE BigData Cup 2024, from preprocessing to training and evaluation.

STUDENT ORGANISATION & VOLUNTEERING

NEECIST (Student Organisation) 2020 – 2024
Board Member Lisbon, Portugal
- Organised **NEECathon**, a student-led hackathon, and ran workshops on Machine Learning and IoT.

Food Bank Against Hunger (Banco Alimentar Contra a Fome) 2014 – Present
Volunteer Portugal

SKILLS & LANGUAGES

Programming: Python, C/C++, TypeScript, Java, SQL, R
AI & Data: RAG, prompt engineering, LLM APIs (OpenAI, Gemini), PyTorch, Hugging Face, computer vision (YOLO)
HPC & Quantum: CUDA, MPI, OpenMP, SIMD, SLURM, Intel VTune, Valgrind, Qiskit Aer, CUDA-Q, Qulacs
Integration & Tools: SAP IDoc, Unified Namespace, HighByte, webhooks, REST APIs, Docker Compose, Git, Linux
Control & Front End: MATLAB/Simulink, PID tuning, RISC-V Assembly, React, Figma
Languages: Portuguese (native), English (C1, Cambridge CAE), Spanish (basic), Dutch (basic)